

25 Astronomy and cosmology

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Stellar luminosity is emitted power; observed flux falls as $L/(4\pi d^2)$ for isotropic emission. A black-body star's peak wavelength decreases as temperature rises according to Wien's law. Cosmological redshift is evidence of expansion; for relatively nearby galaxies, $v \approx H_0 d$, with limitations at very large distances.

Worked example: follow the reasoning

If an identical star is three times farther away, its received flux is one ninth as large. Wien's law $\lambda_{\text{max}} T = 2.90 \times 10^{-3} \text{ m K}$.

Quick check · multiple choice

A If distance to a star doubles, its flux becomes what fraction?

[1]

A One quarter

B One half

C Two times

D Four times

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 A stellar spectrum peaks at 500 nm. Estimate surface temperature with Wien constant $2.90 \times 10^{-3} \text{ m K}$.

[2]

Final answer: _____

Answers and explanation

Quick check: A. One quarter

Question 1 [2 marks]

$$T = 2.90 \times 10^{-3} / (500 \times 10^{-9}) [1] = 5800 \text{ K} [1].$$

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.